Software Systems

Lectures Week 9

Unit 5 - Basic Software Development Techniques

GNU Tools + Modular programming

Part A

Debugging using GDB

Readings: http://www.tutorialspoint.com/gnu_debugger/

Bug

A software bug is an error, flaw, mistake, failure, or fault in a computer program that prevents it from working as intended, producing an incorrect result.

- Bugs can exist at different levels
  - Design
  - Source Code
- Bugs have severities
  - Some bugs produce an incorrect answer
  - Some bugs simply crash an application.
  - Some bugs cause loss of data.
  - Some bugs cause loss of money.
- The worst bugs cause loss of life.

Well known bugs

- Y2K – date overflow
- Ariane 5 Flight 501 – conversion overflow
- MIM-104 Patriot bug – clock drift
- MARS orbiter crash - metric and imperial

Debugging

Debugging is the act of finding the source of a bug and fixing it.

- The hardest part of debugging is finding the problem.
  - This becomes exponentially difficult when the source is very large.
- Just analyzing the code is not always enough to find bugs.
  - You need to run the application.
- Debuggers are tools that help the debugging process.

Debuggers

- Debuggers allow a programmer to run the application in a different mode.
- When running under this different mode ("debug mode"), many new features are available to the programmer.
  - If an application crashes, the programmer can see what line caused the fatal operation. The programmer can also consult the content of the memory.
  - A programmer can analyse an application, line-by-line. At each line, the programmer can consult the content of the memory.
- Most programming languages have a debugger.

To Printf or Not to Printf

- Printf is useful when debugging
  - How can we use it? Suggestions?
- A symbolic debugger can do things printf() can't.
  - halt the program temporarily,
  - list source code,
  - print the datatype of a variable
  - jump to an arbitrary line of code
- You can use a symbolic debugger ON a process that has already crashed and died.

Introduction to GDB

GDB is a debugger which is part of the Free Software Foundation's GNU operating system (eventually just borrowed the Linux kernel…).

GDB can be used to debug:

- C, C++, Objective-C, Fortran, Java(not recent versions) and Assembly programs.
- jdb -> similar utility for Java

Using GDB

First:

- gcc -g -o HelloWorld HelloWorld.c
  - The program is then compiled with additional information such as the source code and symbol table.

This additional information is needed by the debugger.

Starting GDB

Second:

```
gdb HelloWorld
```

The executable

```
GNU gdb Red Hat Linux (5.3post-0.20021129.18rh)
  Copyright 2003 Free Software Foundation, Inc.
  GDB is free software, covered by the GNU General
  Public License, and you are welcome to change it
  and/or distribute copies of it under certain
  conditions. Type "show copying" to see the
  conditions. There is absolutely no warranty for
  GDB. Type "show warranty" for details. This GDB
  was configured as "i386-redhat-linux-gnu"...

(gdb)
```

The command line

Getting Help

Type Help:

```
(gdb) help
```

You can also get help on a specific command by typing “help” and the name of that command:

```
(gdb) help breakpoints
 Making program stop at certain points.
```

Looking at the Source code

You can use the *list* command:

- List
- List filename
- List linenumber
- List function

```
(gdb) list main.c:10
5
6       int main (int argc, char **argv) {
7
8               library* mylibrary = createLibrary(20);
9
10              loadLibrary("lib.txt", mylibrary);
11
12              addBookToLibrary(mylibrary, createBook("Lotr", "Tolkien", 300));
13              addBookToLibrary(mylibrary, createBook("Harry_Potter", "Rowing",
    50));
14              addBookToLibrary(mylibrary, createBook("C_Prog", "Kerning", 100));
```

GNU Debugger with Core Dump

```
$ gcc -g file.c
$ gdb a.out core-dump
  GDB is a free software and you are welcome
   to distribute copies of it under centerain conditions;
   GDB 4.15.2-96q3; Copywrite 2000 Free Software Foundation,
   Inc.

   Program terminated with signal 7, Emulation trap.
   #0 0x2734 in swap (l=93643, u=93864, strat=1) at file.c:110
   110     x=y;

   (gdb) run
```

The run-time error message (crash)

Some Terminology

```
Bash-prompt $ gdb a.out
(gdb) break 4          ← line number (can also be function names, etc.)
(gdb) run              ← run until end, crash, or breakpoint

int main() {
  int age;
  printf("Enter age: ");
  scanf("%d", &age);
                                                  Breakpoint
  if (age > 17)
          printf("Welcome\n");
  else
          printf("Try again when you are older\n");

  return 0;
}
```

Stop & prompt
Step & inspect
Step & inspect
Continue

How to use GDB

- Run you program to see error in Bash
- Use printf to solve problem – not good?
- Take notice to where the problem happened
- Bash-Prompt $ gdb a.out
- (gdb) break before-problem-location
- (gdb) step and print through code to see
- (gdb) change variables, continue or run fn

Step-by-Step

STEP & NEXT

- STEP
  - Step into
  - Go to next line of program. If the next line is a function call then enter into the function.
- NEXT
  - Go to next line of program. If the next line is a function call then do not go into the function just execute the function in its entirety (useful for library functions like printf where we do not have the source code to step into). Go to the next line after the function call.

GDB Command-line Commands

•Quit	end gdb
•List	show 10 lines
•List n,m	show lines n to m
•List function	show all of a function by name
•Run	run your program
•Run (later ctrl-c)	run, then interrupt program
•Run –b < invals > outvals	redirect input and output to program
•Backtrace	see the run-time stack (call stack)
•Whatis x	show x's declaration
•Print x	show value stored at x
•Print fn(y)	execution fn with y as parameter
•Print a @ length	show "length" elements of array a

GDB Command-line Commands

•break LINE_NO	interrupt program at line number
•break FUNC_NAME	interrupt program at function call
•break LINE_NO if EXPR	interrupt at line number if expr true
•break FUNC_NAME if EXPR	interrupt at fn call if expr true
•break FILE_NAME:LINE_NO	interrupt at line number is source-file file
•continue	continue program execution after break
•watch EXPR	stop program as soon as value of EXPR changes
•set variable NAME=VALUE	change contents of a variable
•ptype NAME	pretty print of structure n
•call fn(y)	execute fn with parameter y

GDB Printing Data and History

•(gdb) whatis p
type = int *

•(gdb) print p
$1 = (int *) 0xf8000000

•(gdb) print *p
$2 = Cannot access memory at address 0xf8000000

•(gdb) print $1-1
$3 = (int *) oxf7fffffc

•(gdb) print *$3
$4 = 0

GDB Breakpoint Management

•(gdb) break 17
Breakpoint 1 at 0x2929: file.c, line 17.

•(gdb) break 30 if x == 100
Breakpoint 2 at 0x3550: file.c, line 30.

•(gdb) info breakpoints

Num	Type	Disp	Enabled	Address	What
1	breakpoint	Keep	Y	0x2929	in calc at file.c: 17
2	breakpoint	Keep	Y	0x3550	in sum at file.c: 30

•(gdb) delete 1
•(gdb) delete ← everything!
•(gdb) clear 17 ← any break or watch on line 17
•(gdb) disable n ← do not delete but turn off
•(gdb) enable n
•(gdb) enable once n ← turn on for one time

Where am I?

At any moment, even after a crash, you can use the *where* command to produce a backtrace. (stacktrace)

```
(gdb) where

#0  createBook (title=0x804a218 "Lotr", author=0x804a110
   "Tolkien",
    pages=300) at book.c:8
#1  0x080487fe in loadLibrary (filename=0x8048aa8 "lib.txt",
    myLibrary=0x804a008) at file.c:20
#2  0x08048567 in main () at main.c:10

(gdb)
```

Part B

Modular Programming in C

Readings: Textbook – Section 3.0 Example 9 & 4.0

Modular Programming

A program composed from multiple objects

Other types:

- Single Text File programming (the trivial case)
- Multiple Text File but Single Source programming

Why is modular programming useful?

- Objects are standalone (opens up engineering possibilities)
- Helpful when working in a team of developers
- Helpful when understanding the concept of “named spaces”

Single Source Multiple Text File

We create our program using many .c files, but we merge them at compile-time into a single document.

```
Bash-prompt $ vi file1.c
Bash-prompt $ vi file2.c
Bash-prompt $ vi file3.c
Bash-prompt $ gcc file1.c
Bash-prompt $ ./a.out
```

Notice:
- Three .c files are created, but
- We only compile the first source file
- There exists a command in the first source file that merges the other source files into itself

Benefits:
- More than one person can work on project.

Drawback:
- Compiling entire project each time.

#include

The #include directive has two formats:

- #include<library>
  - Merge a library into your program
- #include "path/textfile"
  - Merge a text file into your program

```
#include "file2.c"                // loads source file from current directory

#include "/user/jack/file2.c"     // loads from specified directory

#include "../jack/file2.c"        // loads from a relative directory
```

Example

In general not a good practice to directly include another C file

Linkedlist.c

```
#include<stdio.h>
void printList(struct NODE *ptr) {
            .....
}
```

MyMax.c

```
#include<stdlib.h>
int max(int a, int b, int c) {
    int aMax = a;
    aMax = (b>aMax) ? b : aMax;
    aMax = (c>aMax) ? c : aMax;
    return aMax;
}
```

Main.c

```
#include<stdio.h>
#include "Linkedlist.c"
#include "MyMax.c"

int main() {
   int a,b,c,result;
   scanf("%d %d %d", &a, &b, &c);
   result = max(a,b,c);
   printf("%d\n", result);
}
```

```
Bash-prompt $ vi Linkedlist.c
Bash-prompt $ vi MyMax.c
Bash-prompt $ vi Main.c
Bash-prompt $ gcc Main.c
Bash-prompt $ ./a.out
```

Modular Programming

We create our program using many .c files.

We compile each source file separately, and merge later.

Notice:

- The gcc –c command compiles each source file separately saving each .c file as a compiled .o file. These .o files cannot execute on their own.
- The gcc command by itself merges all the .o files into an a.out executable file.

```
Bash-prompt $ vi file1.c
Bash-prompt $ vi file2.c
Bash-prompt $ vi file3.c
Bash-prompt $ gcc –c file1.c file2.c file3.c
Bash-prompt $ gcc file1.o file2.o file3.o
Bash-prompt $ ./a.out
```

Only one of them can have a ”main”. Often requires header files to “know” the functions, etc., available in the other C files.

Benefits:
- More than one person can work on project.
- Only changed files needs to be compiled.
- Each file has its own named space.

Drawback:
- Compiling is more complicated.

Named Space

- A semi-private space.
  - Each .o file's global variables and functions are, by default, local to only the .o file it was **defined** & **compiled** within.

File1.c

```
int x;
void printX() {
    printf("%d\n", x);
}
```

File2.c

```
int main() {
    printf("%d\n", x);          // error
    printX();                   // error
}
```

Bash-prompt $ gcc –c File1.c File2.c
Bash-prompt $ gcc File1.o File2.o
Bash-prompt $ ./a.out

Named Space

- Use function prototypes to let the compiler know a function is defined elsewhere.
- Using the extern command a program can access a semi-private variable.

A promise to compiler that printX() and x is defined else where. We will later see a more elegant way of doing this using header files.

File1.c

```
int x;
void printX() {
    printf("%d\n", x);
}
```

File2.c

```
extern int x;
void printX();

int main() {
    printf("%d\n", x);          // legal
    printX();                   // legal
}
```

```
Bash-prompt $ gcc –c File1.c File2.c
Bash-prompt $ gcc File1.o File2.o
Bash-prompt $ ./a.out
```

Example 1 - extern

Linkedlist.c

```
#include<stdio.h>
struct NODE *head;
void printList(struct NODE *ptr) {
            .....
}
```

MyMax.c

```
#include<stdlib.h>
int max(int a, int b, int c) {
    int aMax = a;
    aMax = (b>aMax) ? b : aMax;
    aMax = (c>aMax) ? c : aMax;
    return aMax;
}
```

Main.c

```
#include<stdio.h>
extern struct NODE *head;
void printList(struct NODE *ptr);
int max(int a, int b, int c);

int main() {
    int a,b,c,result;
    scanf("%d %d %d", &a, &b, &c);
    result = max(a,b,c);
    printf("%d\n", result);
    printList(head);
}
```

```
Bash-prompt $ vi Linkedlist.c
Bash-prompt $ vi MyMax.c
Bash-prompt $ vi Main.c
Bash-prompt $ gcc –c Main.c Linkedlist.c MyMax.c
Bash-prompt $ gcc Main.o Linkedlist.o MyMax.o
Bash-prompt $ ./a.out
```

The .h File

Developers sharing semi-private information with other developers without verbal communication. (or choosing not to share)

In the last example we saw, example:

```
extern struct NODE *head;
max();
printList();
```

Type definitions, extern, function prototypes, and #define can be placed in a .h file

Example

Node.h

```
struct NODE {
   int data;
   struct NODE *next;
};
```

Linkedlist.c

```
#include<stdio.h>
#include "Node.h"

struct NODE *head;
void printList(struct NODE *ptr) {
            .....
}
```

Main.c

```
#include<stdio.h>
#include "Node.h"
#include "Max.h"

extern struct NODE *head;

int main() {
   int a,b,c,result;
   scanf("%d %d %d", &a, &b, &c);
   result = max(a,b,c);
   printf("%d\n", result);
}
```

This is considered better form because you avoid duplication errors by writing the definition once.

Example 2

Node.h

```
struct NODE {
   int data;
   struct NODE *next;
};
extern struct NODE *head;
void printList(struct NODE *p);
```

Linkedlist.c

```
#include<stdio.h>
#include "Node.h"

struct NODE *head;
void printList(struct NODE *ptr) {
            .....
}
```

Main.c

```
#include<stdio.h>
#include "Node.h"
#include "Max.h"

int main() {
   int a,b,c,result;
   scanf("%d %d %d", &a, &b, &c);
   result = max(a,b,c);
   printf("%d\n", result);
}
```

This is a more complete example of a .h file.
gcc main.c max.o Linkedlist.o

Problem

Create a project with the following files:

- Main.c
  - Create an account and person
  - Add money and display balance
- Bank.c and Bank.h
  - Array of 10 bank accounts: person ID & balance
  - Balance() withdraw() deposit() createAcct()
- Person.c and Person.h
  - Array of 10 people: name, age
  - AddPerson(), getPersonID()

Problem

Abigail, Bethany, and Sebastian want to work together on a telephone book program. A text file contains the names and numbers in the book (assume one name and number per line). The size of the text file is unknown. A user wants to enter the name and see the number.

How can they work together to create this C program (given what we have covered)?

The Pre-processor

Before gcc compiles your code it will pre-process it.

- This means it will make changes to your source file before it compiles it.

Common pre-processor commands used by developers:

- #include ← we have seen this already
- #define
- #ifdef
- #ifndef

(many other exist, beyond scope of the course)

#define

A technique by which a developer can define new terms or commands that are not part of the C language.

Syntax:

- #define TERM EXPRESSION
  - The EXPRESSION will be inserted into the source code everywhere it finds TERM

Examples:

- #define TRUE 1
- #define FALSE 0
- if (x == TRUE) { … }

Using #define as a macro

A variable #define command

Syntax:

- #define TERM(ARG1, ARG2) EXPRESSION
  - The EXPRESSION will be inserted into the source code everywhere it finds TERM
  - ARG1 and ARG2 will be inserted into the EXPRESSION

Example:

- #define MAX(A,B) (A>B)?A:B
- int result = MAX(5,10); // int result = (5>10)?5:10;

#ifdef and #ifndef

The pre-processor if-statement

Syntax:

- #ifdef TERM
  - If TERM has been previously #define'd, then true, else false.
- #ifndef TERM
  - If TERM has not been previously #define'd, then true, else false.

General syntax:

```
#ifdef TERM
    // multiple C programming statements go here
#else
    // optional #else part, multiple C programming statements go here
#endif
```

Compiling Example

```
#define FRENCH
 // #define ENGLISH
 int main() {
  int age;
  #ifdef FRENCH
      printf("Votre age: ");
  #endif
  #ifdef ENGLISH
      printf("Your age: ");
  #endif
  scanf("%d", &age);
  :
 }
```

This actually compiles your program into a French version without any English.

Instead of keeping two versions of your program you can keep one version and just compile into the language you want.

Crash proof .h file example

Node.h

```
#ifndef NODEH
#define NODEH

 struct NODE {
    int data;
    struct NODE *name;
 };

#endif
```

Often the developer who uses a library does not know all the inner workings of the library (or shared file) and can easily cause a situation where they include a file twice. #ifndef can help.

Lib.h

```
#include "Node.h"
 void addNode(struct NODE * ptr) {
              :
              :
 }
```

```
#include <stdio.h>
#include "Node.h"
#include "Lib.h"

 void main() {
     :
     :
 }
```

Part C

Modular Compiling using make

Readings: chapter 4.1,
http://www.cs.colby.edu/maxwell/courses/tutorials/maketutor/ and
https://sourceware.org/binutils/docs/gprof/

The make Tool

Modular programming:

- Great for team programming
- Fast way to compile large projects

The GNU **make** tool helps us compile complicated projects.

The **make** tool can also do other software engineering activities, like backups, etc.

The make Tool

Composed of:

- The **make** program
- The **makefile** text file

Developers define their software engineering rules within the text file named **makefile**.

The **make** program uses that text file.

If your favorite IDE uses the term **Project**, then inside somewhere is a makefile and a make program.

Basic makefile Text File

```
program: file1.o file2.o
  gcc –o program file1.o file2.o
file1.o: file1.c
  gcc –c file1.c
file2.o: file2.c
  gcc –c file2.c
backup:
  cp *.c /home/project/backup
```

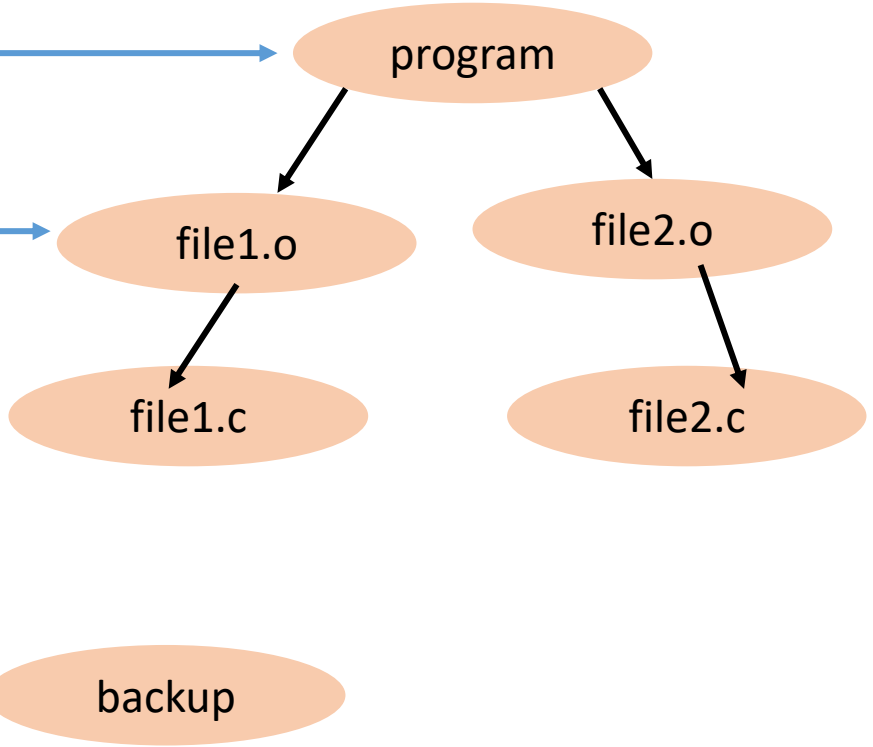
Each label (filename) specifies the other labels it is dependent on, and the command to execute.

Some power exists in selectively executing portions of the makefile.

Basic makefile Text File

Dependency Tree

```
program: file1.o file2.o
    gcc –o program file1.o file2.o
file1.o: file1.c
    gcc –c file1.c
file2.o: file2.c
    gcc –c file2.c
backup:
    cp *.c /home/project/backup
```



Basic makefile Text File

```
program: file1.o file2.o
  gcc –o program file1.o file2.o
file1.o: file1.c
  gcc –c file1.c
file2.o: file2.c
  gcc –c file2.c
backup:
  cp *.c /home/project/backup
```

Method 1:

- Bash-prompt $ make

This invokes the make program default.

It assumes a makefile exists in the same directory and executes the first line. In our example: program: file1.o file2.o

Notice the recursive definition.

Basic makefile Text File

```
program: file1.o file2.o
  gcc –o program file1.o file2.o
file1.o: file1.c
  gcc –c file1.c
file2.o: file2.c
  gcc –c file2.c
backup:
  cp *.c /home/project/backup
```

Method 2:

- Bash-prompt $ make backup

This invokes the make program and specifically targets a portion of the makefile.

It assumes a makefile exists in the same directory and in our example executes: cp *.c /home/project/backup

Notice it executes like Bash.

Demo

Creating the make file for a two file non-recursive factorial program:

- Main.c with the main function
- Factorial.c with the factorial function

More make

How to write a make file that builds multiple executables?

```
all: program1 program2

program1: program1.o …
    gcc -o program1 …
..
program2: program2.o …
    gcc -o program2 …
…
```

Part D

Code Repositories (GIT) – Several others exist (CVS, SVN, Mercurial, …)

Readings: https://www.learnenough.com/git-tutorial?gclid=Cj0KEQjw8tbHBRC6rLS024qYjtEBEiQA7wIDeXo_owx61WwsbBiFPTFsgvcLriD526qlAmford4B1IgaAk9j8P8HAQ

Backup

A folder that contains a copy of your file.

Often the copy is older than the original file because as you develop you modify the original file.

Bash-prompt $ vi file.c

Bash-prompt $ cp file.c backup

Bash-prompt $ vi file.c

If something bad happens to your original file you can always revert back to your backup copy.

Repository

A database containing all the versions of your file (version control system)

This is an improvement over a backup because a repository contains the entire history of all your backup files, also:

- It can be shared
- You can assign permissions
- You can enforce development and deployment rules

Repository History

File1.c version 1 ← Known as the **root** version

↓

File1.c version 2

↓

File1.c version 3

↓

File1.c version 4

↓

File1.c version 5 ← Known as the **tip** version

The files in your working directory are known as the **current** version

Git

A popular version control system

Git was created by Linus Torvalds in 2005 for development of the Linux kernel, with other kernel developers contributing to its initial development. Its current maintainer since 2005 is Junio Hamano.

Git

Working Alone

Should we keep using the backup directory?

How to use basic Git

First – create the folder for your project

Bash-prompt $ mkdir project

Second – put/create the files for the project

Bash-prompt $ cd project

Bash-prompt $ cp ../afile (copy from another dir)

Bash-prompt $ vi anotherfile (you create a new file)

Third – create the Git repository

Bash-prompt $ git init

How to use basic Git

Fourth – Select the files to put in the repo

Bash-prompt $ git add file1 file2

Fifth – Now put the selected files into the repo

Bash-prompt $ git commit –m “message”

- It is very important to add a message describing the changes you made since the last commit. In the future if you will want to undo what you did these messages will help you figure out how far back you may want to go.

Sixth – Done

- This is the basic loop.
- You can now continue working on your files until the next commit, where you will repeat steps 4 and 5.

How to get something back

There are two cases where you may want to undo your work:

- You did a commit but you want to undo it
  - Bash-prompt $ git checkout -f
- You made an error in your original file that you cannot correct, so you want to go back to a previous version
  - Bash-prompt $ git log ← to see commit history, find <SHA>
  - Bash-prompt $ git checkout <SHA>

Which file to commit

Conflicts

There are two cases:

- You were not paying attention to the files you changed and now you would like to commit
  - Bash-prompt $ git diff ← all tip files compared with current
- Your team member worked on the file at the same time as you!!
  - Using Bash:
    - Bash-prompt $ diff yourfile friendfile
    - Bash-prompt $ vi files and copy past, then use git add + git commit
  - Using Git:
    - Bash-prompt $ git diff filename ← compares tip file with current file

Important

Git tries its best to merge different versions of a file automatically by itself.

It will resort to showing you the diff of two or more files when it is having trouble merging.

Git

Working in a Team

Team etiquette: Pull, Commit, Push

The “master” branch and the “official version”

How to make & use a shared Git

Step 1 – create project folder & cd into it

Step 2:

- Start your own shared Git
  - Bash-prompt $ git init --bare ← makes it more compatible
  - Share your URL with friends/team
- Join an existing shared Git
  - Bash-prompt $ git clone <url>
  - Use git add to

The <url> is:

- ssh://user@server/path/folder
- http://user@server/path/folder

Often you will need to ask the system administrator for the URL info & permission.

How to make & use a shared Git

Step 3 – Before you work on any file, make sure that no one else has remotely changed the file

Bash-prompt $ git pull <url>

Step 4 – Now vi your files

Step 5 – Commit to the shared repo

Bash-prompt $ git add <filename>

Bash-prompt $ git commit –m “message”

Bash-prompt $ git push <url>

Git based repositories

- Github https://github.com/
  - git clone https://github.com/<owneruserid>/<reponame>.git
- Git Lab https://gitlab.com/
  - git clone https://gitlab.com/<owneruserid>/<reponame>.git

Others

- Bit bucket
- Source Forge
- …

Summary

Private Repo

Public Repo

Your PC

Local repo

Shared PC

Shared repo

Git init
Git add
Git commit

Git init -bare

Local repo

Person 1 PC

Local repo

Person 2 PC

Git clone
Git pull
Git add
Git commit
Git push

Git clone
Git pull
Git add
Git commit
Git push

How to work in a team

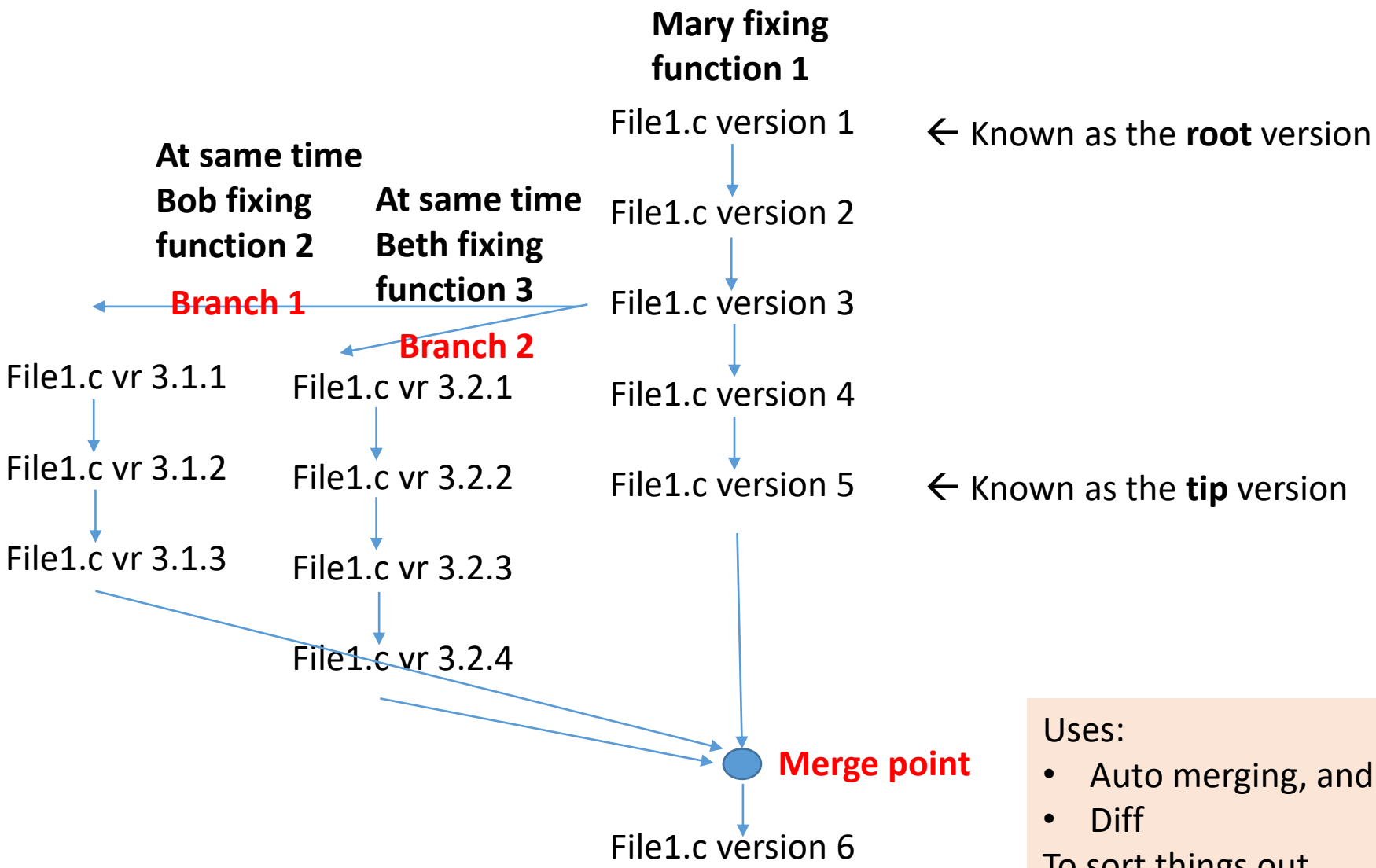
Option 1:

- Assign a source file to a developer.
- No other developer is permitted to edit that file.
- Option 1 can be carried out with the git commands we have already seen.

Option 2:

- Anyone can edit any file.
- Divide the work into branches.
- Option 2 requires us to understand branching.

Repository History



Uses:
- Auto merging, and
- Diff

To sort things out.

How to use branching

Step 1 – See the current branches

Bash-prompt $ git branch

Step 2:

- Join a branch
  - Bash-prompt $ git checkout <branch_name>
- Create your own branch
  - Bash-prompt $ git branch <new_branch_name>
  - Bash-prompt $ git checkout <new_branch_name>
  - Or do it in one shot:
    - Bash-prompt $ git checkout –b <new_branch_name>

Step 3 – now whatever you do it will effect only the branch

How to use branching

Step 5 – To exit a branch

Bash-prompt $ git checkout master

Step 6 – To merge a branch with the master

Bash-prompt $ git checkout master

Bash-prompt $ git merge <branch_name>

Step 7 – Manager branches

- Delete merged branch
  - Bash-prompt $ git branch -d <branch_name>
- Delete a branch that has not been merged
  - Bash-prompt $ git branch -D <branch_name>